Safety Enhanced UAV path planning using spherical vector-based Particle Swarm Optimisation

Group: s2_g18_wc

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Table 1: Team Members

Question 1: **System Objective and Deterministic Baseline**

Project Overview

The objective of this project is to compute a **safe, feasible, and efficient flight trajectory for an unmanned aerial vehicle (UAV)** operating in a three-dimensional terrain environment containing obstacles and elevation variations. The UAV must travel from a predefined start location to a target destination while satisfying several operational constraints, including obstacle avoidance, safe altitude limits, and smooth motion requirements. The optimization framework seeks to **minimize a composite path cost function** that accounts for path length, threat proximity, altitude safety, and maneuver smoothness.

Formally, the optimization problem can be written as

$$\min C = b_1 J_1 + b_2 J_2 + b_3 J_3 + b_4 J_4$$

where

- J_1 represents the path length cost,

- J_2 represents the obstacle avoidance penalty,
- J_3 represents altitude safety deviation,
- J_4 represents path smoothness penalties, and
- b_1, b_2, b_3, b_4 are weighting coefficients.

The goal of the algorithm is therefore to identify a trajectory that minimizes the above cost while respecting the geometric and operational constraints of the UAV.

Deterministic Baseline Strategy

The system has used a deterministic type of baseline, which is Spherical Vector-based Particle Swarm Optimization (SPSO) for unmanned aerial vehicle (UAV) path planning. The model describes a UAV maneuver as a set of intermediate nodes, each having parameters of spherical movement: step length (r), elevation angle (θ), and azimuth angle (ϕ). All these parameters describe the motion of the node relative to its neighbor(s), and all candidate paths satisfy realistic flight dynamics requirements.

A swarm consists of an incarnation of every particle, which is a potential trajectory. During every iteration, the particles change their motion parameters based on deterministic PSO update equations:

$$v_i = w\, v_i + c_1\, r_1\, (Best_i - x_i) + c_2\, r_2\, (gBest - x_i)$$

$$x_i = x_i + v_i$$

where

(a) x_i is the particle position,

(b) v_i is the particle velocity,

(c) $pBest_i$ is the particle's personal best solution,

(d) $gBest$ is the global best solution,

(e) w is the inertia weight, and

(f) c_1, c_2 are learning coefficients.

The swarm considers a wide variety of possible paths, but the rule of decision implies that, in the course of calculations, an algorithm will sample to the direction at the lowest cost value. Consequently, the resulting paths are optimum efficient with all constraints imposed met.

The methodology behind the development of this approach is common to UAV path planning since swarm-based optimization methods are useful in solving nonlinear, high-dimensional optimization that make use of many constraints and high density obstacle environments. The strategy is an essential component in modern UAV navigation studies.

Question 2: **Random Variables in the Simulation**

Random Variable 1: Step Length

Definition

The step length r_i is the span that the UAV covers within each successive node in the planned track.

$$r_i \in [r_{\min}, r_{\max}]$$

Distribution

$$r_i \sim \mathcal{U}(r_{\min}, r_{\max})$$

Where U represents a uniform distribution.

Role in the Simulation

The step length defines the UAV movement in every step of increment on the construction of candidate trajectories. Steps that are longer produce longer segments and those that are shorter allow the creation of smaller adjustments.

Random Variable 2: Angle of elevation.

Definition

The degree of flight ψ_i defines the specific direction that the motion was the UAV with the horizontal plane.

$$\psi_i \in [-\theta, \theta]$$

where θ represents the highest angle of climb in which it is possible.

Distribution

$$\psi_i \sim \mathcal{U}(-\theta, \theta)$$

Role in the Simulation

This variable dictates the vertical aspect of the UAV path, hence having an impact on the variations in altitude when creating the flight path.

Random Variable 3: Azimuth Angle

Definition

The horizontal turning direction of the UAV is recorded in the form of the azimuth angle ϕ_i.

$$\phi_i \in [\phi_{\min}, \phi_{\max}]$$

Distribution

$$\phi_i \sim \mathcal{U}(\phi_0 - \theta_{\max}, \phi_0 + \theta_{\max})$$

Role in the Simulation

The parameter determines the horizontal direction of the movement of the UAV and, therefore, the direction of turning determined during the generation of the trajectory.

Question 3: **Probabilistic Modeling and Fixed Decision Rule** Probabilistic Assumptions

The stochasticity is added by randomly choosing the positions of particles in initialisation, the incorporation of random coefficients in velocity updates and the uniform sampling of the spherical motion variables. The stochastic components allow the algorithm to probe adequately a large solution space.

Deterministic Decision rule

Each time the algorithm runs the cost function is evaluated at all candidate trajectories at every iteration. The deterministic rule consists in choosing the path that reduces the costs of the path. Under this, the algorithm is determined to select a set of trajectories, which realize the minimum aggregate cost in a deterministic manner.

$$x^* = \arg\min_x C(x)$$

Path Construction Mathematical Representation.

Spherical parameters of motion are used to generate the UAV trajectory and this transformation gives us the spherical motion vectors in Cartesian coordinates which is to be used in a future cost assessment.

$$x_i = x_{i-1} + r_i \cos(\psi_i) \sin(\phi_i) \tag{1}$$
$$y_i = y_{i-1} + r_i \cos(\psi_i) \cos(\phi_i) \tag{2}$$
$$z_i = z_{i-1} + r_i \sin(\psi_i) \tag{3}$$

Question 4: **Simulation Pipeline**

Stage 1: Input Generation

The environment model is opened by loading:

- terrain elevation data
- obstacle locations
- start position
- destination location
- search boundaries

The following parameters determine the field of operation of the UAV.

Stage 2: Particle Pre-Initialization.

A pool of candidate solutions is obtained. Parameters of each particle are spherical motion parameters, which indicate a possible UAV trajectory. Particles are randomly started at achievable positions.

Stage 3: Path Construction

The coordinate transformation equations transform the spherical parameters into cartesian coordinates, thus generating the real UAV trajectory in space in three dimensions.

Stage 4: Deterministic Cost Evaluation.

Every candidate path is assessed with the help of a cost function, which consists of four elements:

- path length
- obstacle avoidance
- altitude constraint
- trajectory smoothness

The cumulative weighted sum total of these elements dictates the aggregate path cost.

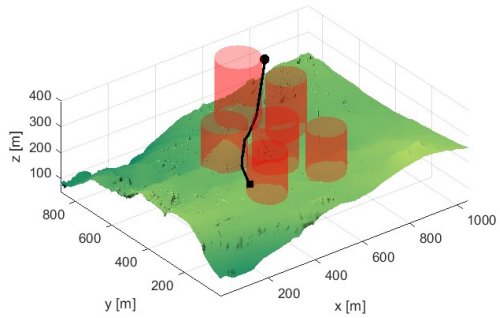
Stage 5: Swarm Optimization

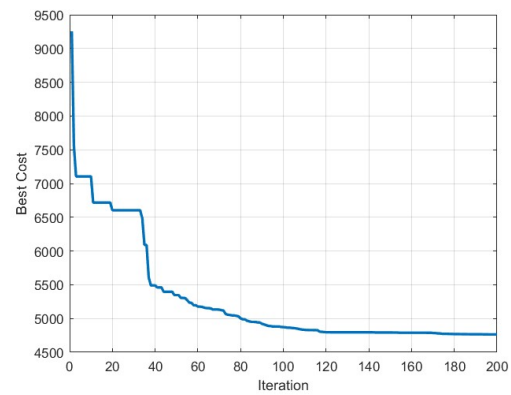
The SPSO velocity update rule is used to update the position of particles. Unless cost improvements are made in a globally best solution, personal best and global best solutions are improved as an improvement in cost occurs.

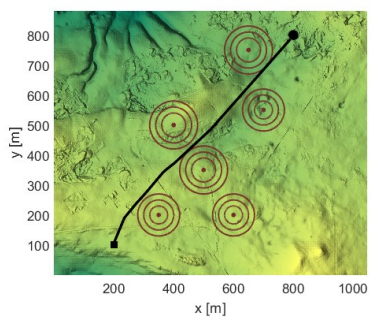
Stage 6: Convergence and Output.

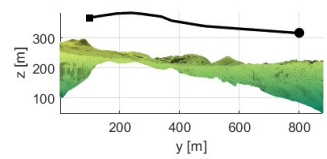
The algorithm will have:

- the optimal UAV trajectory
- a convergence curve which is a representation of the cost reduction as one iterates.









Output Diagrams

Question 5: **Simulation Outputs and Interpretation**

Performance Metrics

The SPSO-based UAV path-planning system performance becomes tested by the weighted cost-based weighted cost-function that consolidates various criteria of qualities of the trajectories:

$$C = b_1 J_1 + b_2 J_2 + b_3 J_3 + b_4 J_4$$

This is because there is a performance measure that is related to each term.

Path Length ((J_1))

This measure is used to measure the overall range covered by the UAV:

$$J_1 = \sum_{i=1}^{N-1} |P_{i+1} - P_i|$$

The short routes result in the minimization of travel time and energy usage.

Obstacle Avoidance ((J$_2$))

Routes that come close to cylindrical hindrances are punished. Minimal Euclidean distance between each path segment with the centre of an obstacle is calculated and a penalty is created whenever the UAV is in one of a danger zone.

Altitude Constraint ((J$_3$))

This is to make sure that the UAV does not go out of safe altitude with respect to terrain elevation. Great punishments are inflicted in the case of UAV falling below the ground level.

Path Smoothness ((J$_4$))

Turns that are sharp or sudden changes in the UAV climb behaviour are penalised to maintain physically realistic UAV motion.

The solution finally followed by the algorithm is that which reduces the total cost by a minimal degree.

Convergence Analysis

Convergence analysis applies to the SPSO algorithm whereby it is analyzed to reveal that the path solution is refined with each subsequent iteration. The algorithm stores the optimal cost indicator within every iteration:

$$\text{BestCost}(t)$$

at which the iteration number of t is used.

A convergence curve is a curve that represents the iteration number versus the best cost. The cost is usually characterized by a steep drop in the initial iterations as the particles search the search space, and it then levels off as the swarm gathers around an optimal path.

To measure algorithmic stability, several simulations can be conducted, and the end cost values analyzed with the help of the plots of PDF and CDF. A small distribution with a variance of low values indicates good convergence and quality of the paths.

On the whole, the convergence behavior shows that SPSO makes reasonable tradeoffs between exploration and exploitation to result in safe and efficient UAV trajectories.

Observed Result

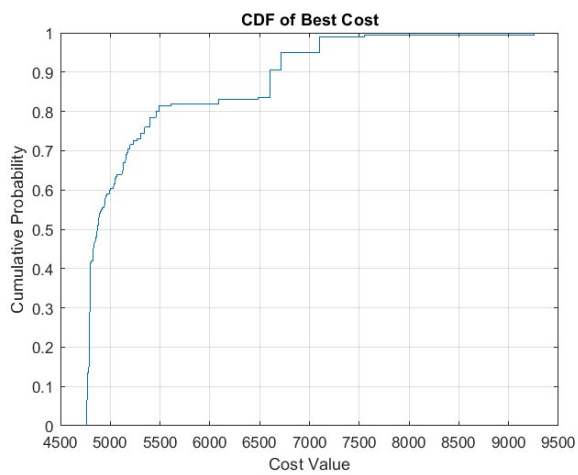


Figure 1: CDF of localization error

Question 6: **Technical Gaps and Preparation for Next Stage**
The simplified Modeling Assumptions.

Assumptions That have been put in the current model include:

- static environment,
- fixed position of obstacles,
- ideal terrain data,
- reduced UAV traditional motion constraints, limited only to turning and climb constraints.

These assumptions are also useful in simplification of the simulation, but might not be appropriate to conditions in the real-world context of operations.

Current Constraints of the Deterministic Baseline.

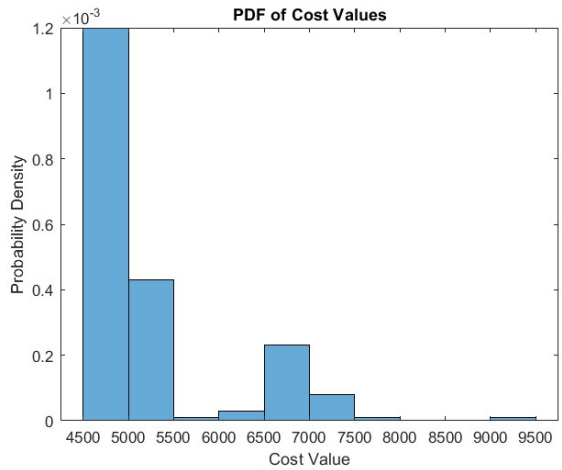


Figure 2: PDF of localization error

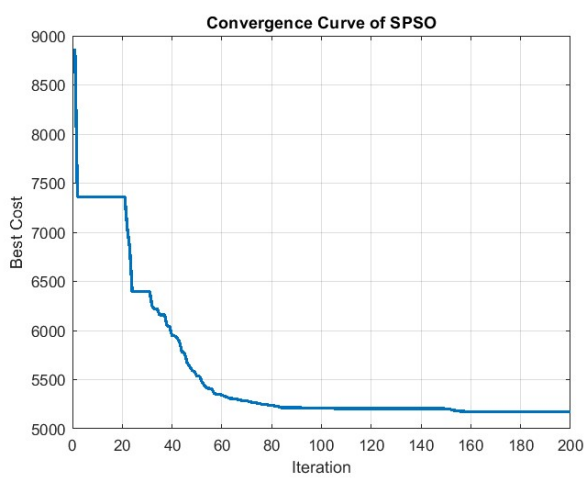


Figure 3: Convergence Curve of SPSO

Optimisation of geometric paths is the key concern of the SPSO base although it does not introduce the downstream features: - ground based UAV communication interactions, wind and environmental disturbance,

- dynamic or moving obstacles,
- power use in flying of the UAV.

Potential Future Work Role of Randomness.

Subsequent stages can involve probabilistic modelling to better represent real world uncertainties and these are:

- disruption of the environment,
- sensor imprecision,
- positions of uncertain obstacles.

It is anticipated that the inclusion of stochastic elements in planning the trajectories of UAVs would increase robustness and reliability.

Milestone 4 Improvements Scheduled.

The future improvements could include the following:

- refining the costing model by ensuring that it reflects the flight restrictions,
- tolerance to the unpredictable input circumstances or the adversarial substance,
- test against algorithmic stability by multipurpose simulation,
- study of simple extensions to enhance the reliability of paths.